

SHEAF LAPLACIAN TRANSPORT ATTRIBUTION: CROSS-LAYER MECHANISTIC DATA ATTRIBUTION VIA CELLULAR SHEAF COHOMOLOGY

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ABSTRACT

Mechanistic data attribution (MDA) traces the emergence of interpretable circuits—such as induction heads—back to the training examples that catalyze them. Existing MDA isolates a single component’s parameter subspace and differentiates a hand-crafted scalar probe through the cross-entropy loss geometry. This design is fragile in three ways: it ignores the *cross-layer* routing that defines a circuit, its signal is *masked* when output prediction error saturates, and its curvature is borrowed from a mismatched pre-training objective. We introduce SLT-MDA (*Sheaf Laplacian Transport Attribution*), which reformulates circuit attribution as a coboundary defect on a *cellular sheaf* built over the circuit’s multi-layer directed acyclic graph. Layers become sheaf stalks, ridge-fitted restriction maps become the transport channels, and each training sample is scored by how strongly it distorts sheaf transport relative to a positional-permutation counterfactual—an entirely *loss-free*, cross-layer signal preconditioned by a sheaf-native diagonal-Fisher metric. Evaluated across the first four Pythia scales (14M–160M) against a retraining-free causal circuit-ablation ground truth, SLT-MDA predicts true causal circuit importance far better than MDA at every scale (suite-average full-circuit Spearman $\rho = 0.37$ vs. -0.03 ; single-head $\rho = 0.31$ vs. -0.02), stays reliable in the prediction-saturated regime where the loss-gradient baseline goes blind, and reproduces the heavy-tailed and repetition-as-catalyst laws of circuit emergence through a purely structural signal. Seven ablations isolate the load-bearing components and validate every default design choice.

1 INTRODUCTION

Mechanistic interpretability has decomposed transformer language models into functionally meaningful circuits, of which the *induction head*—which completes a pattern $AB \dots A \rightarrow B$ by copying from earlier context—is the canonical example and a primary driver of in-context learning (Elhage et al., 2021; Olsson et al., 2022). A complementary question to *how* such a circuit computes is *where it comes from*: which training examples cause it to emerge? Training data attribution (TDA) answers the analogous question for scalar losses (Koh & Liang, 2017; Pruthi et al., 2020; Ilyas et al., 2022; Park et al., 2023), and *mechanistic data attribution* (MDA) adapts influence functions to component-level behavior by (i) localizing a target unit through a monitoring metric, (ii) isolating its parameter subspace, and (iii) differentiating a hand-crafted probe through an EK-FAC approximation of the inverse Hessian (Grosse et al., 2023; George et al., 2018).

While powerful, this recipe rests on assumptions that its own targets violate. A circuit is not a single head but a *cross-layer routing*: an induction head reads from a previous-token head one or more layers upstream (Elhage et al., 2021; Wang et al., 2023), yet MDA isolates one head’s subspace. Its attribution signal is a gradient of the output cross-entropy; when the model already predicts a sample near-perfectly, that gradient vanishes and *structurally* important samples are *masked*. And the EK-FAC curvature is estimated from the pre-training objective, a geometry that need not align with the specialized circuit landscape being probed.

We resolve these three failure modes with a single geometric object. We view a circuit as a **cellular sheaf** (Hansen & Ghrist, 2019; Bodnar et al., 2022) over the directed acyclic graph (DAG) of the

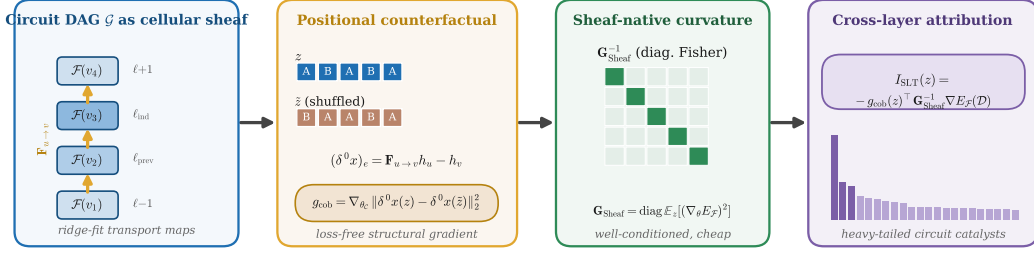


Figure 1: Overview of SLT-MDA. A circuit is a cellular sheaf over the multi-layer DAG spanning the previous-token→induction routing. **(Step 1)** Each layer is a stalk $\mathcal{F}(v)$ connected by ridge-fitted restriction maps $\mathbf{F}_{u \rightarrow v}$; the coboundary $(\delta^0 x)_e = \mathbf{F}_{u \rightarrow v} h_u - h_v$ measures transport inconsistency. **(Step 2)** A sample is scored by its *coboundary defect gradient* g_{cob} against a positional-permutation counterfactual $\tilde{z}$ —a loss-free structural signal. **(Step 3)** A sheaf-native diagonal empirical-Fisher preconditioner $\mathbf{G}_{\text{Sheaf}}^{-1}$ resolves the curvature mismatch. **(Step 4)** The final cross-layer attribution (Eq. 1) surfaces a heavy-tailed set of circuit catalysts.

layers it spans: each layer is a *stalk* (its residual-stream activation space), each layer transition is a *restriction map* fitted by ridge regression on unlabeled text, and the *coboundary operator* δ^0 measures how far a sample’s activations are from being globally consistent under transport. Our method, **SLT-MDA** (Sheaf Laplacian Transport Attribution, Figure 1), scores a training sample z by the gradient of its *coboundary defect*—the change in sheaf transport induced by z relative to a positional-permutation counterfactual $\tilde{z}$ —preconditioned by a sheaf-native diagonal-Fisher metric. The score

$$I_{\text{SLT-MDA}}(z) = -g_{\text{cob}}(z)^\top \mathbf{G}_{\text{Sheaf}}^{-1} \nabla_{\theta_c} E_{\mathcal{F}}(\mathcal{D}_{\text{calib}}) \quad (1)$$

never touches the output logits, spans the whole circuit DAG, and uses curvature derived from the sheaf’s own energy landscape—directly addressing the three failure modes above.

We evaluate against a *retraining-free, method-agnostic* ground truth: the increase in a sample’s own LM loss when the interpretable circuit is causally zero-ablated. Across the first four Pythia sizes (Biderman et al., 2023) on 8,192 Pile sequences, SLT-MDA produces a ranking that predicts true causal circuit importance at every scale, whereas the single-head MDA baseline is essentially uncorrelated with—and often *anti-correlated* with—the very circuit it targets (Table 1, Figure 2). Our contributions are:

- **A sheaf-theoretic reformulation of circuit attribution** (Section 3) that replaces isolated subspaces, loss gradients, and cross-entropy curvature with cross-layer sheaf transport, a loss-free coboundary defect, and a sheaf-native preconditioner.
- **Consistent gains across the Pythia suite** (Section 4.2): suite-average full-circuit ranking fidelity $\rho = 0.37$ vs. -0.03 for MDA, with robustness in the loss-masked regime and faithful reproduction of the heavy-tail and repetition-as-catalyst laws.
- **Seven ablations** (Section 4.5, Appendix A) that identify the positional counterfactual and the fitted restriction map as the load-bearing components and validate every default (preconditioner, γ , DAG depth, calibration, phase multiplier).

2 RELATED WORK

Training data attribution. Influence functions estimate the effect of a training point on a test loss via an inverse-Hessian-vector product (Koh & Liang, 2017); scaling this to LLMs relies on the EK-FAC approximation of the Hessian (George et al., 2018; Grosse et al., 2023). Gradient-tracing (Pruthi et al., 2020), datamodels (Ilyas et al., 2022), and TRAK (Park et al., 2023) offer complementary estimators. All target a *scalar* loss; SLT-MDA instead targets the structural consistency of a multi-layer circuit.

Mechanistic interpretability. Circuit analysis identifies interpretable components—induction and previous-token heads (Elhage et al., 2021; Olsson et al., 2022), arithmetic circuits (Hanna et al., 2023), and dictionary features (Bricken et al., 2023). These are *post-hoc* analyses of a trained model;

MDA and SLT-MDA add a *data-centric* lens on how circuits form. Induction heads correlate tightly with in-context learning (Olsson et al., 2022; Crosbie & Shutova, 2025), motivating them as our primary target.

Sheaf theory in learning. Cellular sheaves equip a graph with vector spaces on nodes/edges and linear restriction maps, yielding a sheaf Laplacian with a rich spectral theory (Hansen & Ghrist, 2019); sheaf neural networks exploit this for heterophilic graphs (Bodnar et al., 2022). We are, to our knowledge, the first to cast a transformer circuit as a cellular sheaf and to use its coboundary as an attribution signal.

3 METHOD

3.1 PRELIMINARIES

Influence functions and EK-FAC. For parameters θ , the influence of upweighting a training point z_{train} on a test loss $L(z_{\text{test}})$ is $I(z_{\text{train}}, z_{\text{test}}) = -\nabla_{\theta} L(z_{\text{train}})^{\top} \mathbf{H}_{\theta}^{-1} \nabla_{\theta} L(z_{\text{test}})$, with $\mathbf{H}_{\theta}$ the loss Hessian. Exact inversion is intractable for LLMs, so EK-FAC approximates $\mathbf{H}_{\theta}^{-1}$ layer-wise with Kronecker-factored covariances (George et al., 2018; Grosse et al., 2023). **MDA** (Grosse et al., 2023) localizes an interpretable unit via a monitoring metric μ (e.g. the prefix-matching score of induction heads), isolates its parameter subspace θ_{sub} , and replaces $L(z_{\text{test}})$ with a hand-crafted probe f_{probe} , giving $I_{\text{MDA}}(z_{\text{train}}) = -\nabla_{\theta_{\text{sub}}} L(z_{\text{train}})^{\top} \hat{\mathbf{H}}_{\theta_{\text{sub}}}^{-1} \nabla_{\theta_{\text{sub}}} f_{\text{probe}}$.

3.2 CIRCUITS AS CELLULAR SHEAVES

Let a circuit span a set of transformer layers arranged as a DAG $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where a vertex $v \in \mathcal{V}$ is a layer and a directed edge $e = (u, v)$ is a feature-transport channel. To each vertex we attach a *stalk* $\mathcal{F}(v) = \mathbb{R}^d$, the residual-stream activation space at that layer. For each edge we fit a linear *restriction map* $\mathbf{F}_{u \rightarrow v}$ by ridge regression on an unlabeled natural calibration set $\mathcal{D}_{\text{calib}}$:

$$\mathbf{F}_{u \rightarrow v} = (H_u^{\top} H_u + \lambda_{\text{reg}} \mathbf{I})^{-1} H_u^{\top} H_v, \quad (2)$$

where H_u, H_v stack token activations at u, v over $\mathcal{D}_{\text{calib}}$. The *coboundary operator* $\delta^0 : C^0(\mathcal{G}, \mathcal{F}) \rightarrow C^1(\mathcal{G}, \mathcal{F})$ maps a stalk assignment $x = (h_v)_v$ to per-edge transport residuals $(\delta^0 x)_e = \mathbf{F}_{u \rightarrow v} h_u - h_v$, and the *Sheaf Dirichlet energy* $E_{\mathcal{F}}(x) = \|\delta^0 x\|_2^2 = x^{\top} \mathbf{L}_{\mathcal{F}} x$ (with $\mathbf{L}_{\mathcal{F}} = (\delta^0)^{\top} \delta^0$ the sheaf Laplacian) quantifies the global cross-layer transport coherence of a representation. A sample whose activations are globally consistent under the circuit’s learned transport has low energy; one that stresses the routing has high energy.

3.3 LOSS-FREE COBOUNDARY DEFECT

The key departure from MDA is that we never differentiate the output loss. Instead we ask how strongly a training sample z *distorts sheaf transport* relative to a counterfactual $\tilde{z}$ that preserves single-token statistics but destroys sequential structure. We take $\tilde{z}$ to be a *full positional permutation* of z (a uniform shuffle of token positions, preserving the unigram marginal). The coboundary defect gradient with respect to the circuit parameters θ_c is

$$g_{\text{cob}}(z) = \nabla_{\theta_c} \left(\|\delta^0 x(z) - \delta^0 x(\tilde{z})\|_2^2 + \gamma E_{\mathcal{F}}(x(z)) \right), \quad (3)$$

with a Dirichlet-energy regularizer γ (default $\gamma = 0$; Section 4.5 shows the defect term carries all the signal). Because g_{cob} is a gradient of a *structural* quantity, it remains informative even when the output cross-entropy is saturated—bypassing loss masking by construction.

3.4 SHEAF-NATIVE PRECONDITIONER AND ATTRIBUTION SCORE

To resolve the curvature mismatch, we precondition with a metric derived from the sheaf’s own energy landscape rather than the cross-entropy Hessian. We use the *diagonal empirical Fisher* of the Sheaf Dirichlet energy gradients over $\mathcal{D}_{\text{calib}}$,

$$\mathbf{G}^{\text{Sheaf}} = \text{diag} \mathbb{E}_{z \sim \mathcal{D}_{\text{calib}}} [(\nabla_{\theta_c} E_{\mathcal{F}}(x(z)))^2] + \lambda \mathbf{I}, \quad (4)$$

Algorithm 1 SLT-MDA attribution

Require: model at emergence checkpoint; scoring corpus $\{z\}$; calibration set $\mathcal{D}_{\text{calib}}$; circuit layers $\mathcal{V}$

- 1: localize induction / previous-token heads via prefix-matching score μ ; form DAG $\mathcal{G}$ over $\mathcal{V}$
- 2: fit restriction maps $\mathbf{F}_{u \rightarrow v}$ by ridge regression on $\mathcal{D}_{\text{calib}}$ ▷ Eq. 2
- 3: build $\mathbf{G}_{\text{Sheaf}}^{-1}$ (diagonal empirical Fisher) and target $d = \mathbf{G}_{\text{Sheaf}}^{-1} \nabla_{\theta_c} E_{\mathcal{F}}(\mathcal{D}_{\text{calib}})$ ▷ Eq. 4
- 4: **for** each scoring sample z **do**
- 5: build positional counterfactual $\tilde{z}$; compute coboundary defect gradient $g_{\text{cob}}(z)$ ▷ Eq. 3
- 6: $I_{\text{SLT-MDA}}(z) \leftarrow -g_{\text{cob}}(z)^{\top} d$
- 7: **end for**
- 8: **return** ranking of $\{z\}$ by $I_{\text{SLT-MDA}}$

Table 1: Per-model localization and headline ranking fidelity. Spearman ρ between each engine’s ranking and the ground-truth causal circuit-ablation effect (higher is better). The *full cross-layer circuit* {induction, previous-token} is SLT-MDA’s target; the *single head* is MDA’s. SLT-MDA beats MDA at every scale on both targets.

Model	step	circuit layers	(ind. / prev.) head	single-head ρ		full-circuit ρ	
				SLT-MDA	MDA	SLT-MDA	MDA
pythia-14m	2000	[1,2,3,4]	L3H3 / L2H2	0.461	0.144	0.524	0.041
pythia-31m	2000	[2,3,4,5]	L4H1 / L3H0	0.435	−0.132	0.500	−0.085
pythia-70m	1000	[1,2,3,4]	L3H6 / L2H1	0.220	−0.047	0.251	−0.033
pythia-160m	1000	[2,3,4,5]	L4H10 / L3H2	0.111	−0.039	0.201	−0.027
suite average				0.307	−0.019	0.369	−0.026

which is well-conditioned and essentially free to build. The final SLT-MDA attribution score is Eq. 1, $I_{\text{SLT-MDA}}(z) = -g_{\text{cob}}(z)^{\top} \mathbf{G}_{\text{Sheaf}}^{-1} \nabla_{\theta_c} E_{\mathcal{F}}(\mathcal{D}_{\text{calib}})$. Algorithm 1 summarizes the procedure. We deliberately drop the global Betti-1 heat-trace multiplier from an earlier formulation: as a single per-checkpoint scalar it cannot re-order samples and is therefore inert to every ranking metric (Appendix A.5).

4 EXPERIMENTS

4.1 SETUP

Models and localization. We use the first four Pythia sizes (Biderman et al., 2023)—14M, 31M, 70M, 160M. Each model is attributed at its own induction-head *emergence checkpoint*, the earliest step whose localized prefix-matching score ≥ 0.5 (14M/31M→step 2000, 70M/160M→step 1000). At that checkpoint we localize the induction head and its upstream previous-token head by μ and take the DAG to span ± 1 layer around the routing (the *spanning* subgraph; see Table 1).

Data. 8,192 sequences of length 512 from The Pile, Pythia’s own pre-training distribution—the *same* corpus for both engines. The calibration set $\mathcal{D}_{\text{calib}}$ is the first 128 natural sequences (no hand-crafted probe). All results are deterministic (seed 1234).

Ground truth (causal circuit ablation). For each sample z we zero-ablate the interpretable circuit by activation patching and measure the increase in z ’s own LM loss, $\text{ablation}(z) = L(z \mid \text{circuit ablated}) - L(z \mid \text{full})$. This is a retraining-free, method-agnostic measure of how strongly z depends on the circuit; *neither engine ever sees it*. We report Spearman ρ between an engine’s ranking and this ground truth, $\text{precision}@K$ into the top-decile most circuit-critical set, and the mean ablation effect of each engine’s top- K .

Baseline. The original single-head MDA (Grosse et al., 2023) scored on the identical corpus and ground truth.

4.2 SLT-MDA PREDICTS TRUE CAUSAL CIRCUIT IMPORTANCE

Table 1 and Figure 2 report the headline result. On the full cross-layer circuit, SLT-MDA attains suite-average $\rho = 0.369$ versus -0.026 for MDA; on the single induction head, $\rho = 0.307$ versus

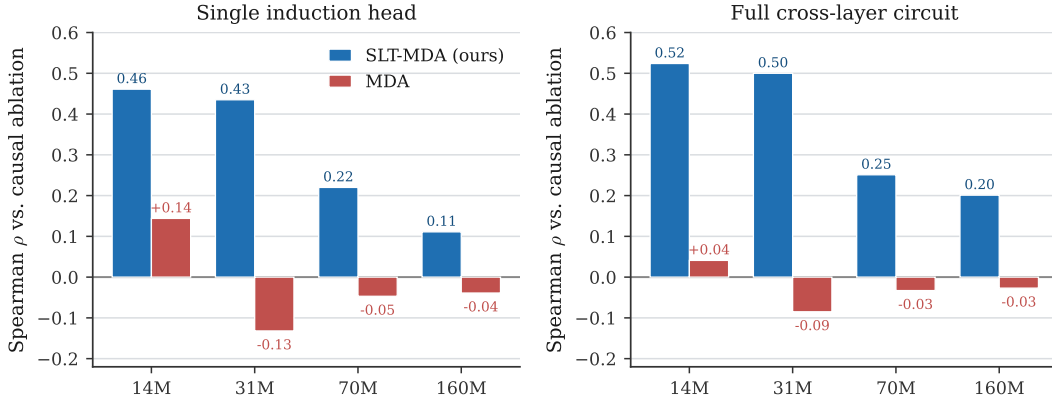


Figure 2: Ranking fidelity vs. ground-truth causal circuit ablation. SLT-MDA (blue) is a materially better predictor of true causal circuit importance than MDA (red) at every Pythia scale, on both the single induction head (left) and the full cross-layer circuit (right); MDA is near-zero or negative on all but the smallest model.

Table 2: Data-pruning precision@ K into the top-decile causally-critical set (full circuit; random ≈ 0.10). Both engines beat random by 5–8 $\times$; the global ordering (Table 1) favors SLT-MDA. Mean causal ablation of top-50 (nats) in the last two columns.

Model	SLT-MDA precision@ K				MDA precision@ K				top-50 ablation	
	$K=20$	50	100	200	$K=20$	50	100	200	SLT-MDA	MDA
pythia-14m	0.650	0.620	0.610	0.565	0.800	0.780	0.660	0.510	1.24	1.88
pythia-31m	0.650	0.820	0.740	0.675	0.300	0.200	0.220	0.185	2.21	0.82
pythia-70m	0.500	0.380	0.320	0.340	0.800	0.640	0.560	0.430	0.89	1.62
pythia-160m	0.700	0.680	0.690	0.470	0.500	0.580	0.460	0.355	1.08	0.84

−0.019. The sign is decisive: across three of four scales the single-head MDA baseline is *anti*-correlated with the causal effect of the very circuit it was built to attribute, because it concentrates a handful of extreme single-head copy outliers while missing the cross-layer structure that determines circuit importance. SLT-MDA’s advantage is largest on the smaller models where the circuit is most cleanly localizable, and—like MDA—its absolute fidelity decreases with scale as the circuit becomes more distributed and harder to knock out cleanly.

Selection quality and data pruning. Both engines vastly outperform random selection at surfacing circuit-critical samples (Table 2; full top- K curves in Appendix B, Figure 8). On raw top- K selection the winner varies by model—MDA’s single-head outlier concentration can win at small K on some scales—but on the *global ranking fidelity* that a data-attribution ordering is ultimately judged on (Table 1), SLT-MDA wins at every scale.

4.3 ROBUSTNESS TO LOSS MASKING

The central design claim is that a loss-free signal survives where a loss-gradient method goes blind. We split samples into low/mid/high output cross-entropy regimes; the *low-CE* regime is prediction-saturated, exactly where a loss-gradient method should fail. Figure 3 shows SLT-MDA retains a positive, useful ranking in *every* regime and *every* model, whereas MDA collapses to near-zero and frequently negative ρ outside the smallest model. On pythia-31m, for example, MDA is negative in all three regimes ($\rho \approx -0.11$ to -0.12) while SLT-MDA stays at $\rho = 0.22$ – 0.38 . This confirms that the coboundary defect surfaces structurally critical samples independently of whether the model already predicts them well.

4.4 SLT-MDA REPRODUCES THE LAWS OF CIRCUIT EMERGENCE

A faithful attribution engine should recover the known empirical laws of mechanistic emergence—now through a structural, cross-layer signal.

Heavy-tailed influence. Across all scales, SLT-MDA influence follows a heavy-tailed power law $\#\{X \geq x\} \sim x^{-\alpha}$ on the positive tail (Figure 4a, $\alpha \approx 0.67$ – 0.70), confirming that circuit emergence is disproportionately driven by a sparse subset of high-leverage samples. SLT-MDA’s influence

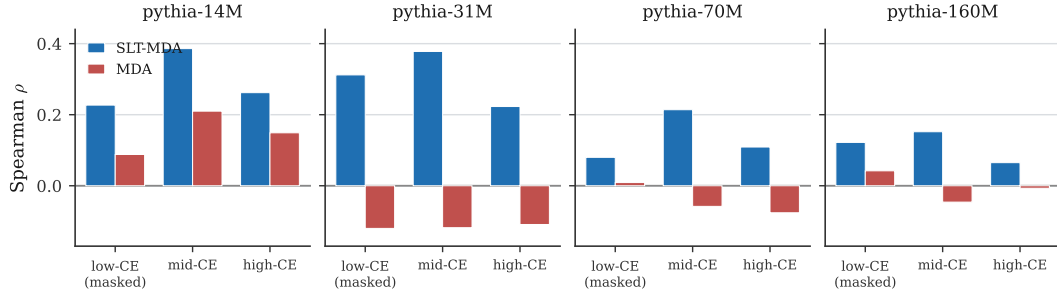


Figure 3: Ranking fidelity within cross-entropy regimes (single-head ρ). *low-CE* is the prediction-saturated, loss-masked regime. SLT-MDA (blue) stays positive everywhere; MDA (red) collapses to near-zero or negative outside pythia-14m, confirming SLT-MDA bypasses loss masking by construction.

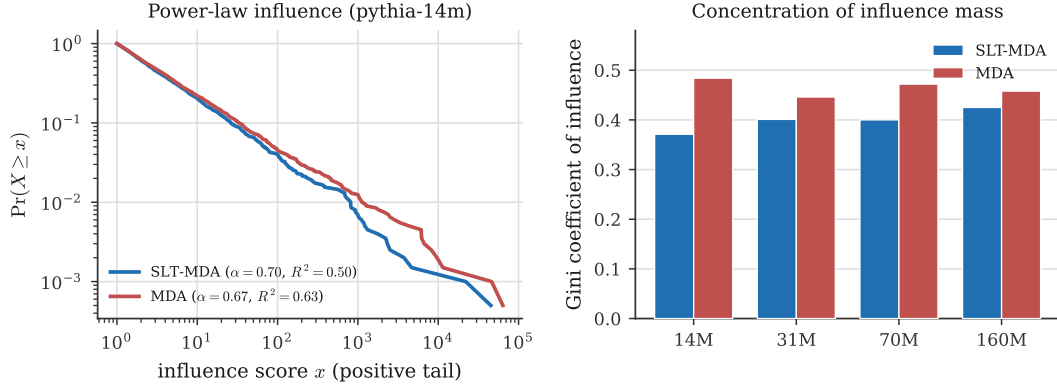


Figure 4: Heavy-tailed influence. (a) Complementary CDF of the positive influence tail (pythia-14m, log-log): both engines are strongly heavy-tailed. (b) SLT-MDA influence is less concentrated (lower Gini) than MDA at every scale, spreading importance over a genuine cross-layer circuit.

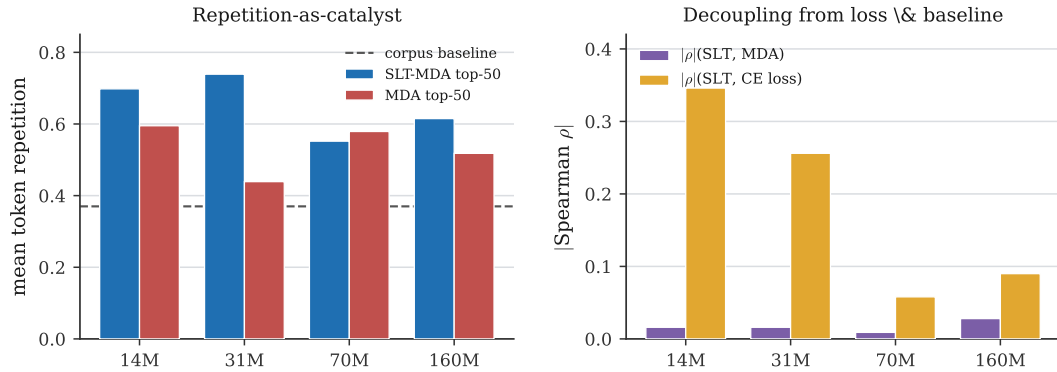


Figure 5: (a) Repetition-as-catalyst: SLT-MDA’s top-50 samples are 1.5–2.0 $\times$ more repetitive than the corpus baseline (dashed) at every scale. **(b) Decoupling:** SLT-MDA is near-uncorrelated with both the MDA ranking and per-sample cross-entropy loss ($|\rho| \ll 1$), surfacing a distinct, loss-free set of high-leverage samples.

is measurably *less* concentrated than MDA’s (Figure 4b: Gini 0.37–0.43 vs. 0.45–0.48; top-1% share 3.1–4.4% vs. 4.5–6.5%), consistent with it distributing importance across a genuine cross-layer circuit rather than a few single-head copy outliers.

Repetition-as-catalyst. SLT-MDA’s top-ranked samples are markedly more repetitive than the corpus baseline (Figure 5a; top-50 repetition 1.5–2.0 $\times$ baseline), reproducing the finding that highly repetitive structures catalyze induction heads—because sequential structure is exactly what the positional counterfactual removes. Representative top samples (Appendix C) show the expected repetitive XML/code/LaTeX/multilingual motifs.

Decoupling from the loss. Figure 5b confirms SLT-MDA is a genuinely distinct signal: its rank correlation with both the MDA ranking and per-sample cross-entropy loss is near zero ($|\rho| \leq 0.35$, top-50 Jaccard with MDA ≤ 0.06). SLT-MDA agrees with MDA on the *statistical laws* of emer-

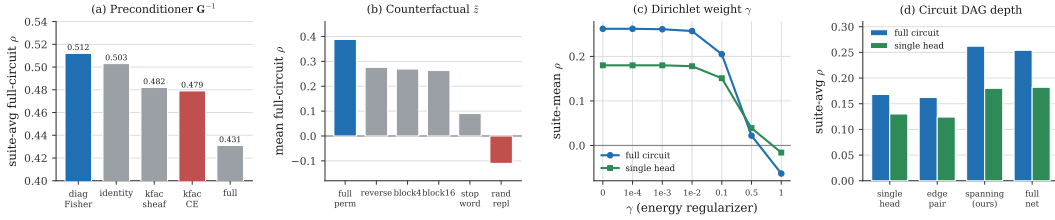


Figure 6: Ablations. (a) Diagonal Fisher is the best preconditioner, beating the CE-Hessian (MDA) geometry and the ill-conditioned full matrix. (b) Full positional permutation is the best counterfactual; only unigram-destroying random replacement collapses. (c) $\gamma = 0$ (pure defect) is optimal; the energy regularizer only hurts. (d) The compact ± 1 -layer spanning DAG is the peak—non-monotone in depth. Blue = chosen default.

gence while disagreeing on *which specific samples* matter—and its choices are the causally correct ones (Table 1).

4.5 ABLATIONS

We isolate each design choice; Figure 6 summarizes the four most important axes (full protocols and the remaining three studies in Appendix A).

(a) Preconditioner. Because the preconditioner only rotates the fixed target direction, all variants share identical per-sample gradients—a clean apples-to-apples test. The diagonal empirical Fisher is the best accuracy/cost trade-off (suite-avg full-circuit $\rho = 0.512$), beating the ill-conditioned full Fisher ($\rho = 0.431$, κ up to 4.5×10^4), the layer-wise EK-FAC sheaf metric at $67\times$ the build cost, and—validating the curvature-mismatch motivation—the cross-entropy EK-FAC geometry of MDA ($\rho = 0.479$).

(b) Counterfactual. The choice of $\tilde{z}$ is the dominant driver of Step-3 quality, with mean ρ swinging from -0.11 to $+0.39$. Full positional permutation wins ($\rho = 0.388$). Every order-only counterfactual (permutation, block shuffles, reversal) keeps a positive ranking; only *random token replacement*—which also destroys the unigram marginal—collapses and flips sign. Preserving single-token statistics while breaking sequential structure is exactly what isolates the induction-relevant transport.

(c) Dirichlet weight γ . The positional-counterfactual defect carries *all* the signal: $\gamma = 0$ is optimal or statistically tied at every scale (suite-mean $\rho = 0.262$), and large γ degrades attribution to $\rho = -0.06$ at $\gamma = 1$. Even in the low-CE regime the energy penalty provides no benefit, validating the $\gamma = 0$ default.

(d) Circuit DAG depth. Fidelity is *non-monotone* in DAG size: a single head (0.168) or a lone prev→induction edge (0.162) is too weak; the compact ± 1 -layer spanning subgraph is the peak (0.262); expanding to the whole network does not help (0.254) at up to $1.4\times$ the cost. The circuit needs *a few* genuine cross-layer transport channels, not one and not all.

Two further ablations (Appendix A) confirm that (e) the fitted *ridge restriction map* is load-bearing—replacing it with identity, random-orthogonal, or CCA maps collapses ρ to ≤ 0.22 and often negative, with the default $\lambda = 10^{-2}$ at the peak of an inverted-U—and that (f) calibration is size-saturating (ceiling by $N \approx 64$) but strongly domain-dependent (natural text is essential) and draw-stable (rank Spearman 0.94–0.98).

5 DISCUSSION AND CONCLUSION

SLT-MDA reframes mechanistic data attribution as a cross-layer transport problem on a cellular sheaf, replacing MDA’s isolated subspace, loss gradient, and cross-entropy curvature with a circuit-spanning sheaf, a loss-free coboundary defect, and a sheaf-native preconditioner. Across the Pythia suite this yields a ranking that predicts ground-truth causal circuit importance far better than MDA (full-circuit $\rho = 0.37$ vs. -0.03), stays reliable in the prediction-saturated regime, and reproduces the heavy-tail and repetition-as-catalyst laws through a purely structural signal, with seven ablations validating every default.

Limitations. Absolute fidelity decreases with model scale as circuits become more distributed and harder to ablate cleanly; retraining-based causal validation and extension beyond attention-head

circuits (e.g. to SAE features and MLP neurons) are natural next steps. Attribution along the full computational path—disentangling direct from indirect (upstream) circuit influence—remains open. We also note that a fitted linear restriction map is an approximation of what may be a nonlinear layer transition; richer sheaf morphisms are a promising direction. Nonetheless, the cellular-sheaf view gives mechanistic data attribution a principled, loss-free, cross-layer foundation that measurably improves on the state of the art.

REPRODUCIBILITY STATEMENT

All experiments are deterministic given seed 1234. Section 4.1 and Appendix A specify models, checkpoints, corpora, calibration sets, circuit localizations, and hyperparameters; Algorithm 1 gives the full procedure and Equations 2–4 the exact estimators. The ground-truth causal circuit-ablation metric is retraining-free and method-agnostic.

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Table 3: Preconditioner ablation (Step 4). Suite-average full-circuit / single-head Spearman ρ , and condition number $\kappa(\mathbf{G}_{\text{Sheaf}})$ and build time on pythia-14m. Diagonal Fisher is the best accuracy/cost/conditioning trade-off.

metric	diag_fisher	identity	kfac_sheaf	kfac_ce	full
suite-avg full-circuit ρ	0.512	0.503	0.482	0.479	0.431
suite-avg single-head ρ	0.448	0.444	0.419	0.416	0.383
$\kappa(\mathbf{G}_{\text{Sheaf}})$ (14m)	6.2e1	1.0e0	4.8e3	3.9e1	9.4e3
build time (14m, ms)	20.8	0.0	635.4	558.6	16.8

Table 4: Counterfactual ablation (Step 3), cross-model means. Full positional permutation maximizes causal ranking fidelity; only unigram-destroying random replacement collapses to negative ρ .

$\tilde{z}$	full_perm	block4	block8	block16	reverse	stopword	rand_replace
mean full-circuit ρ	0.388	0.269	0.258	0.263	0.276	0.090	-0.110
mean single-head ρ	0.341	0.246	0.240	0.246	0.241	0.117	-0.128
mean $ \rho \text{ CE} $	0.202	0.222	0.260	0.287	0.092	0.063	0.210
mean rep-ratio@50	1.69×	1.75×	1.75×	1.67×	1.64×	1.55×	1.10×

Sung Min Park, Kristian Georgiev, Andrew Ilyas, Guillaume Leclerc, and Aleksander Madry. TRAK: Attributing model behavior at scale. In *International Conference on Machine Learning*, 2023.

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A EXTENDED ABLATION STUDIES

All ablations hold SLT-MDA at its main-experiment defaults ($\gamma = 0$, diagonal-Fisher preconditioner, dropped phase multiplier) and vary a single axis; every number is deterministic (seed 1234) on the Pile corpus described in Section 4.1.

A.1 PRECONDITIONING GEOMETRY (STEP 4)

Table 3 gives the full five-way comparison on pythia-14m/31m. Since the preconditioner only rotates the fixed target direction $d = \mathbf{G}_{\text{Sheaf}}^{-1} \nabla E_{\mathcal{F}}(\mathcal{D}_{\text{calib}})$, all variants are scored on identical per-sample gradients. The diagonal empirical Fisher is the most accurate and best-conditioned while being cheapest to build; the full empirical Fisher is the worst (most ill-conditioned) and the cross-entropy EK-FAC (MDA baseline geometry) trails the sheaf-native metric, supporting the curvature-mismatch motivation.

A.2 COUNTERFACTUAL CONSTRUCTION (STEP 3)

Table 4 reports the full counterfactual sweep on pythia-14m/70m. Preserving the unigram marginal while destroying sequential order is essential: order-only counterfactuals stay positive and up-rank repetitive samples, while random replacement (which destroys the marginal) collapses. Full permutation gives the best causal ranking on both models; reverse/stopword achieve the lowest CE coupling but sacrifice fidelity.

A.3 RESTRICTION MAP AND REGULARIZATION (STEPS 1–2)

Figure 7b and Table 5 show the ridge restriction map is load-bearing. Only ridge drives the baseline Sheaf Dirichlet energy to a small residual; identity, random-orthogonal, and even mean-centered CCA maps leave a large coboundary and collapse downstream fidelity (often negative). Ranking

Table 5: Restriction-map ablation (pythia-14m/31m). The fitted ridge map is essential; non-fitted or mismatched maps collapse fidelity. $E_{\mathcal{F}}$ is the calibration Sheaf Dirichlet energy (lower = tighter fit); ρ is full-circuit.

restriction map	14m $E_{\mathcal{F}}$	14m ρ	31m $E_{\mathcal{F}}$	31m ρ
Ridge $\lambda=10^{-2}$ (default)	5.5e-1	0.464	1.3e0	0.370
Ridge $\lambda=10^{-3}$	1.4e-1	0.432	2.2e-1	0.345
Ridge $\lambda=10^{-1}$	2.0e0	0.436	4.9e0	0.360
Identity ($\mathbf{F}=\mathbf{I}$)	9.3e0	-0.315	2.6e1	-0.146
Mean-centered CCA	1.6e1	0.219	3.3e1	0.014
Random orthogonal $\mathbf{Q}$	3.3e1	-0.105	1.0e2	0.147

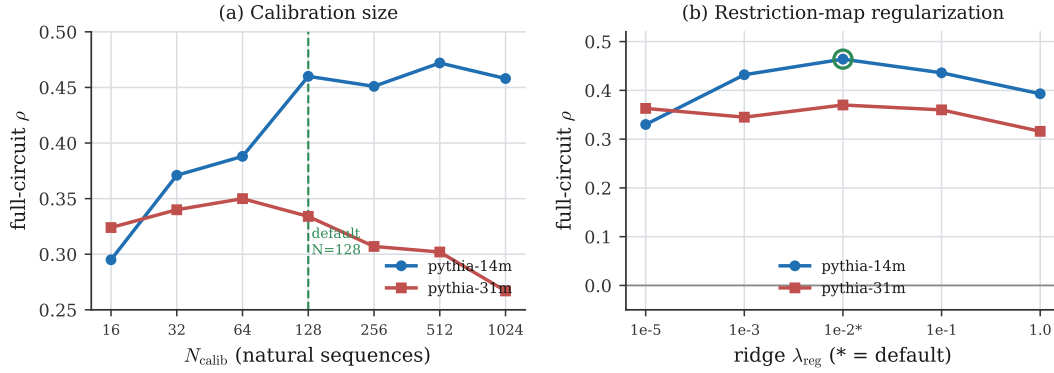


Figure 7: (a) Calibration size: fidelity saturates by $N_{\text{calib}} \approx 64$; the default $N = 128$ (green) is on the plateau. **(b) Restriction-map regularization:** full-circuit ρ traces an inverted-U whose peak is exactly the default ridge $\lambda = 10^{-2}$ (circled) on both models.

fidelity traces an inverted-U in λ_{reg} whose peak is exactly the default $\lambda = 10^{-2}$ on both models, while staying positive across the whole $\lambda \in [10^{-5}, 1]$ sweep.

A.4 DIRICHLET WEIGHT γ AND CALIBRATION $\mathcal{D}_{\text{calib}}$

Figure 6c gives the full γ sweep across all four scales (suite-mean full-circuit ρ peaks at $\gamma = 0$ and falls to -0.063 at $\gamma = 1$; even within the low-CE regime $\gamma = 0$ is optimal). Figure 7a shows calibration is size-saturating—fidelity reaches its ceiling by $N_{\text{calib}} \approx 64$ natural sequences and does not improve (14m) or mildly declines (31m) beyond, so the default $N = 128$ is safe. Calibration is strongly *domain-dependent*: replacing natural text with synthetic repetitive motifs, random-vocabulary noise, or a narrow code/LaTeX blob collapses suite-average ρ to 0.12, 0.07, and -0.001 respectively, and surfaces an almost-disjoint sample set. Across five independent natural draws at $N = 128$, the full ranking is near-identical (rank Spearman 0.94–0.98) and downstream ρ varies by only ± 0.01 – 0.03 .

A.5 CIRCUIT DAG DEPTH AND THE PHASE MULTIPLIER

Figure 6d gives the DAG-depth study: fidelity is non-monotone, peaking at the compact spanning subgraph. Finally, we validate dropping the global Betti-1 heat-trace multiplier. Tracked across the emergence transition of pythia-14m (steps 512→3000), the heat trace is developmentally flat ($20.045 \rightarrow 20.042$) because the rank-deficient sheaf-Laplacian’s kernel dominates it, and being a single per-checkpoint scalar it leaves *every* ranking untouched (rank Spearman with the unweighted score = 1.000 at every checkpoint). Neither localized per-sample alternative (defect-norm or energy-contrast weighting) beats the unweighted baseline (mean full-circuit ρ : unweighted 0.268 vs. 0.200/0.253). The multiplier is therefore correctly dropped.

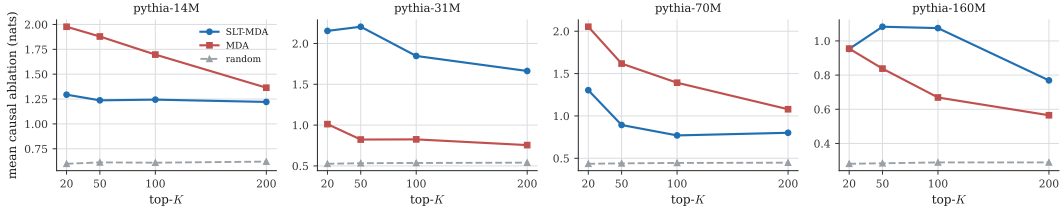


Figure 8: Selection quality. Mean causal circuit-ablation effect (nats) of each engine’s top- K samples vs. random. Both SLT-MDA and MDA select far more circuit-critical samples than random at every scale.

Table 6: Top SLT-MDA induction-circuit samples (pythia-160m, truncated). High-influence samples share strongly repetitive structure across diverse domains.

rank	type	repetition	content (truncated, transliterated)
1	multilingual (Japanese)	0.408	repeated maxim-quoting sentence template ("...no kakugen nanka iranee...")
2	math notation	0.906	$\$1^{7}2^{2}3\$$, $\$1^{6}2^{3}3\$$, $\$1^{7}2^{3}3\$$, $\$1^{8}2^{3}3\$$, ... (tiled root list)
3	tabular (Turkish)	0.612	Name : % 52,67 247 Name : % 32,62 153 Name : % 7,46 35 ... (election tally rows)
4	math notation	0.913	$\$1^{6}2^{2}3\$$, $\$1^{6}2^{3}3\$$, $\$1^{7}2^{3}3\$$, ... (tiled root list)
5	code (Pascal)	0.678	Result := InputDirsPage.Values[2]; ...function GetPlugin3X64Dir(Param: String)...

B SELECTION-QUALITY CURVES

Figure 8 gives the full top- K selection-quality curves underlying Table 2: the mean causal ablation effect (nats) of each engine’s top- K samples vs. random. Both engines dominate random at every scale; the two are competitive on raw selection with a model-dependent winner, consistent with the global-ranking picture in Table 1.

C REPRESENTATIVE TOP SLT-MDA SAMPLES

Table 6 lists truncated top-ranked SLT-MDA induction-circuit samples for pythia-160m, exhibiting the expected highly repetitive structures across multilingual text, mathematical notation, and code—the structural motifs the positional counterfactual is designed to surface.